

Problem Brief

Meridian Forge Robotics · The Process Below

LEARNER

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ARTIFACT

Problem Brief

STATUS

Working document

SECTION 01

Visible Failure

According to Nia Voss, Night QA Supervisor, MR-17 started repeating Test K-44 at about 2:13 a.m., even though K-44 is no longer in use. The incident log supports her account. MR-17 accepted TEST_K44_START at 02:13:07 and started its repeat cycle at 02:13:09. At 02:14:02 the operator sent a stop request, and MR-17 answered with ACTIVE_INSTRUCTION_PENDING. MR-17 received the stop request, but the repeat cycle kept going and the test did not stop.

SECTION 02

Verified Facts

MR-17 accepted TEST_K44_START at 02:13:07. (Incident log, Nia Voss)

The repeat cycle began at 02:13:09. (Incident log, Nia Voss)

The operator stop request at 02:14:02 returned

ACTIVE_INSTRUCTION_PENDING. (Incident log, Nia Voss)

The current scheduler shows no active test assignment for MR-17, and K-44 is not in its test catalog. (Current scheduling console, Records Terminal)

K-44 was marked retired 186 days before the incident and removed from the active workflow. (Maintenance record, maintenance archive)

The command was validly formatted and its source label is legacy_sync.exe. (Command telemetry, Elias Chen)

Production is paused in the affected QA cell. (Staff statements, Mara Ilyan)

SECTION 03

Assumptions and Open Questions

Assumption: MR-17 behaved normally before the incident. (Nia Voss, based only on what she observed; not checked against logs)

Assumption: legacy_sync.exe caused the entire incident. (Telemetry does not prove the broader cause)

Open question: Who owns legacy_sync.exe, and why is it running?

Open question: Was K-44 removed from every copy, service, and schedule? (Maintenance record does not confirm)

Open question: What does MR-17's full software history show? (Mara Ilyan: not yet checked)

SECTION 04

System Environment Map

SECTION 04A

People and Roles

Nia Voss, Night QA Supervisor: observed the incident and keeps the incident records.

The operator: sent the stop request at 02:14:02.

Elias Chen, Robotics Systems Engineer: holds the command telemetry.

Mara Ilyan, Production Operations Manager: manages production in the affected QA cell.

Owner of legacy_sync.exe: unknown.

SECTION 04B

Processes and Rules

Tests are assigned to MR-17 through the current scheduler. Retired tests are removed from the active workflow through maintenance (K-44 retired 186 days before the incident).

Operators can send a stop request to MR-17.

While an instruction is active, a stop request returns

ACTIVE_INSTRUCTION_PENDING and the instruction continues.

SECTION 04C

Information and Data

Incident log:

TEST_K44_START accepted

02:13:07, repeat cycle

02:13:09, stop request

02:14:02.

Current scheduler assignments and test catalog: no assignment for MR-17, no K-44.

Maintenance archive: K-44 retirement record.

Command telemetry: source label legacy_sync.exe.

MR-17 full software history: not yet reviewed.

SECTION 04D

Technology and Tools

MR-17 robot in the QA test cell.

Current scheduling console (Records Terminal).

Internal command interface that delivered the K-44 command.

legacy_sync.exe, a service whose owner and purpose are unverified.

SECTION 04E

Environment and Constraints

Production is paused in the affected QA cell.

The owner and purpose of legacy_sync.exe are unverified.

MR-17's full software history has not been reviewed.

A restart could clear information the investigation still needs (Nia Voss), so the robot is not being operated.

SECTION 05

Investigation Boundary

Inside the Investigation

How the K-44 command reached MR-17: the legacy_sync.exe service, the internal interface it used, and the current scheduler that shows no assignment.

The K-44 retirement and whether it removed every copy, service, and schedule.

How MR-17 handles a stop request while an instruction is active (ACTIVE_INSTRUCTION_PENDING). MR-17's full software history before and during the incident.

Outside for Now

Restarting or operating MR-17: Nia Voss stated the team is not being asked to operate the robot, and a restart could clear information the investigation still needs.

Production scheduling and recovery in the QA cell: this is Mara Ilyan's area, and current evidence does not show it caused the incident.

Other robots and test cells: no record so far shows they received retired tests, so they are not required yet.

SECTION 06

Organizational Impact

Confirmed: production is paused in the affected QA cell (Mara Ilyan, Production Operations Manager). MR-17 ran a retired test in a repeat cycle, and an operator stop request did not halt it (Incident log). Not yet confirmed: the scale of the impact, such as lost hours, units, or cost, has not been reported, and it is unknown whether other robots or schedules can receive retired tests.

SECTION 07

Problem Statement

A test retired 186 days before the incident was still delivered to MR-17 through an internal interface from a source labeled legacy_sync.exe, outside the current scheduler, and once running it could not be interrupted by an operator stop request. As a result, production in the affected QA cell is paused while the source and extent of the retired test's availability remain unverified.

SECTION 08

Next Investigation Step

Action: review the service records for legacy_sync.exe alongside MR-17's full software history. Source: the system owner of legacy_sync.exe and MR-17's software history records. Question it could answer: who owns legacy_sync.exe, why it is still running, and whether it still holds a copy of Test K-44 that can issue commands to MR-17 outside the current scheduler.

SHARED ARTIFACT

Working Document Link

https://github.com/mattyle19782020-netizen/CIT140-Portfolio/blob/87d5c3a1d27d53990816973c47a5ef41dd7b0007/Week01_Something_Isnt_Adding_Up/artifact/Le_Matthew_CIT140_Week01_Problem_Brief.pdf

WEEK 1 STOPPING RULE

Define before designing.

This Problem Brief records the problem that requires investigation. It does not recommend a solution.